Multi Platform Reputation Analysis System

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Abstract

- This project employs deep learning for aspect-based sentiment analysis, gathering and processing data from various platforms to assess product and service reputation.
- Integrating spam detection and precise sentiment analysis, it calculates numerical reputation values for entities and their specific aspects.
- Unlike prior models, this approach ensures comprehensive and accurate evaluations, aiming to enhance the precision and reliability of reputation assessments across diverse online platforms.

Introduction

- In the internet age, evaluating user-generated opinions demands advanced automated systems. Existing methods, focused on reputation scores, often ignore malicious manipulation.
- Our novel approach combines spam filtering and aspect-based sentiment analysis.
- Unlike current systems providing only entity-level scores, our project aims for finer-grained aspect-based assessments.
- Utilizing deep learning algorithms and natural language processing, the goal is to improve accuracy and reliability in assessing product and service reputations across diverse online platforms.

Existing system

The Existing system evaluate the reputation of products, services, or entities across multiple platforms by considering specific aspects or attributes like quality and customer service.

Disadvantages:

Lack the ability to effectively process reviews from various platforms, filter out spam provide nuanced assessments , and offer advanced visualization tools

Proposed system

Our “MultiPlatform Reputation Analysis System" tackles limitations to offer consumers reliable and nuanced reputation information.

- By evaluating opinions across platforms and focusing on specific aspects, it delivers precise insights, meeting the need for robust Multiplatform reputation management and aiding informed decisions on products and services.

Literature survey

S.NO	TITLE	AUTHOR	METHODOLOGY	LIMITATIONS
1	Cross-Platform Reputation Assessment	Smith, J. et al.	Utilized machine learning algorithms for sentiment analysis and incorporated review popularity metrics.	Limited dataset diversity; did not address spam detection or offer aspect-specific reputation evaluations.
2	Aspect-Based Sentiment Analysis for Reviews	Wang, H. et al.	Employed deep learning models for aspect-based sentiment analysis. Considered temporal patterns in reviews.	Lacked integration of spam detection and did not explore aspects beyond sentiment, such as review popularity.
3	Addressing Spam in Reputation Evaluation	Chen, Q. and Kim, M.	Focused on spam detection techniques in online reviews. Used statistical methods to filter out malicious content.	Limited emphasis on sentiment analysis and did not explore the integration of review popularity and posting timing.

System Requirements

Hardware Requirements:

Operating system :windows only

Processor: i5 and above

Ram:8gb and above

Hard disk: 25gb and above

Software Requirements:

Anaconda 3

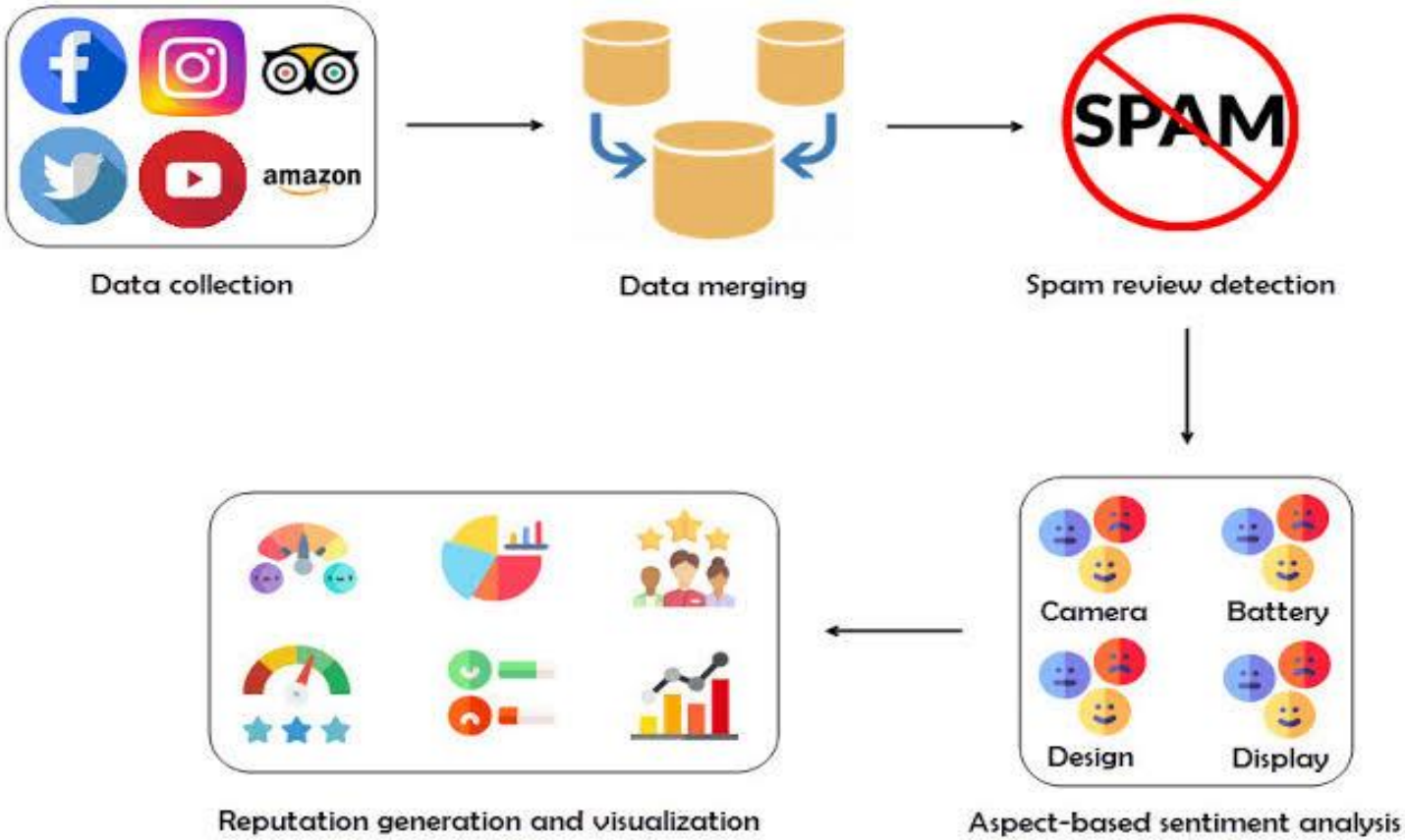
Modules

- **Data Collection module:**
  - This module is responsible for gathering reviews and opinions from various platforms such as Twitter, Face book, Amazon, etc.
- **Spam Filtering module:**
  - This module focuses on identifying and filtering out reviews posted by malicious users or spam accounts.
- **Aspect-Based Sentiment Analysis module:**
  - Analyzes sentiment for each aspect to provide a more granular understanding of user opinions.
  - Utilizes sentiment analysis techniques to determine whether aspects are positively or negatively perceived.

Modules

- **Numerical Reputation Value Computation module:**
  - Combines the results from the previous modules to compute numerical reputation values for the overall entity and its individual aspects.
- **Visualization Tool module:**
  - Develops an advanced visualization tool to display detailed information about the generated reputation values.

Methodology



- **Data collection:**

  Our innovative system uniquely collects and processes data from various online platforms. Through web scraping, we merged details from platforms like Amazon and Twitter, classifying them into two types. NLP techniques were then applied for text cleaning, ensuring a standardized dataset for effective reputation analysis.

- **Spam detection:**

  To counteract opinion spammers attempting to manipulate reputations, our system focuses on two normalized spammer behavioral features. By collecting opinions from various platforms, we prioritize spam detection. While platforms like Amazon and Yelp have made strides in spam review detection, our approach ensures a trustworthy reputation value for informed decision-making by customers.

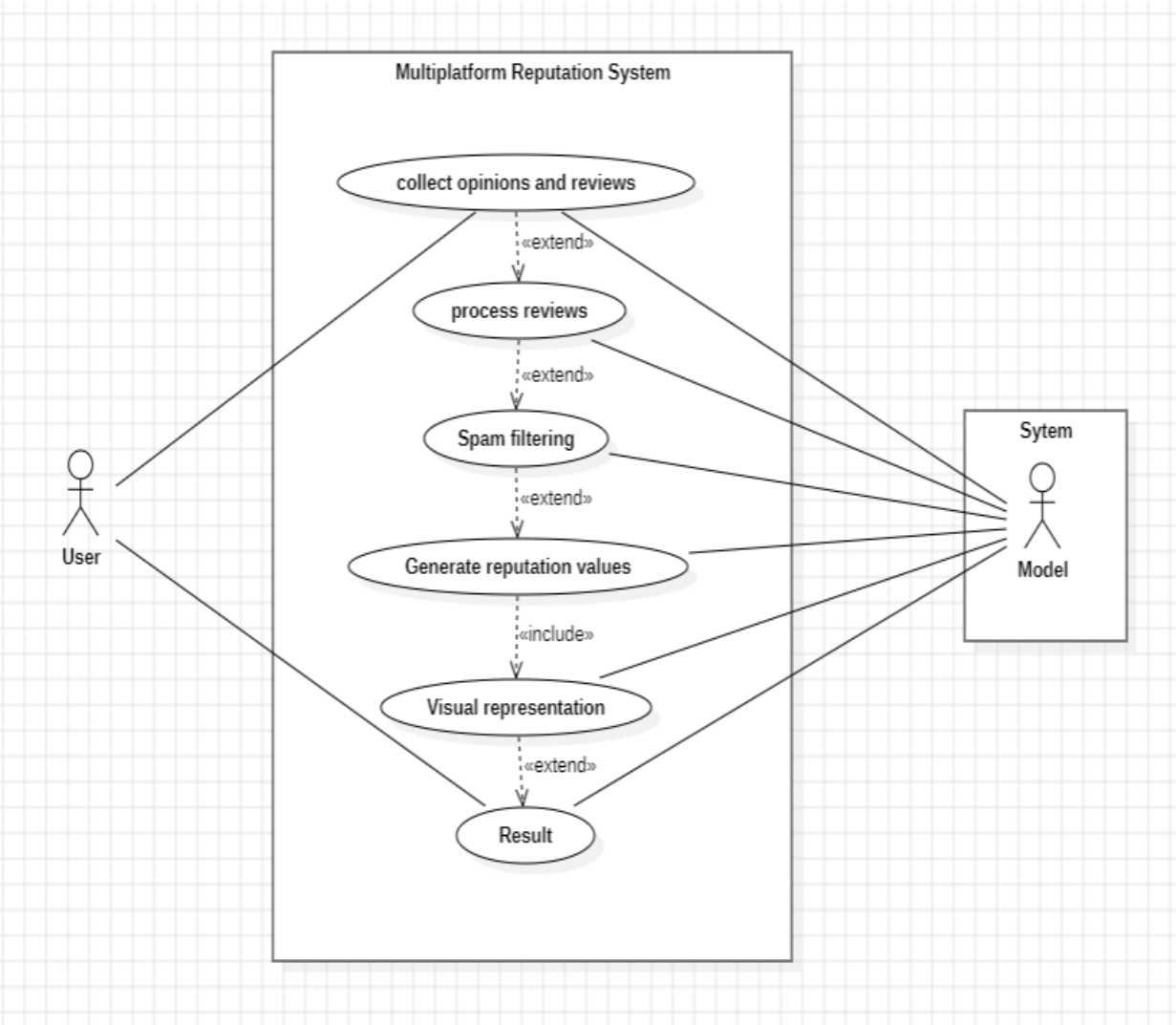
- **Reputation generation and visualization:**

Our system evaluates Internet opinions by factoring in spam detecting, and aspect sentiment orientation to calculate numerical scores, influencing the final entity reputation value. The user-friendly interface offers detailed insights into overall and aspect-specific reputation values, and influential reviews. Enhanced with an interactive visualization tool, users can gain deeper insights by hovering and clicking on specific features. This tool facilitates informed decision-making by providing comprehensive insights into the targeted product or service, surpassing previous reputation systems in efficiency and usability.

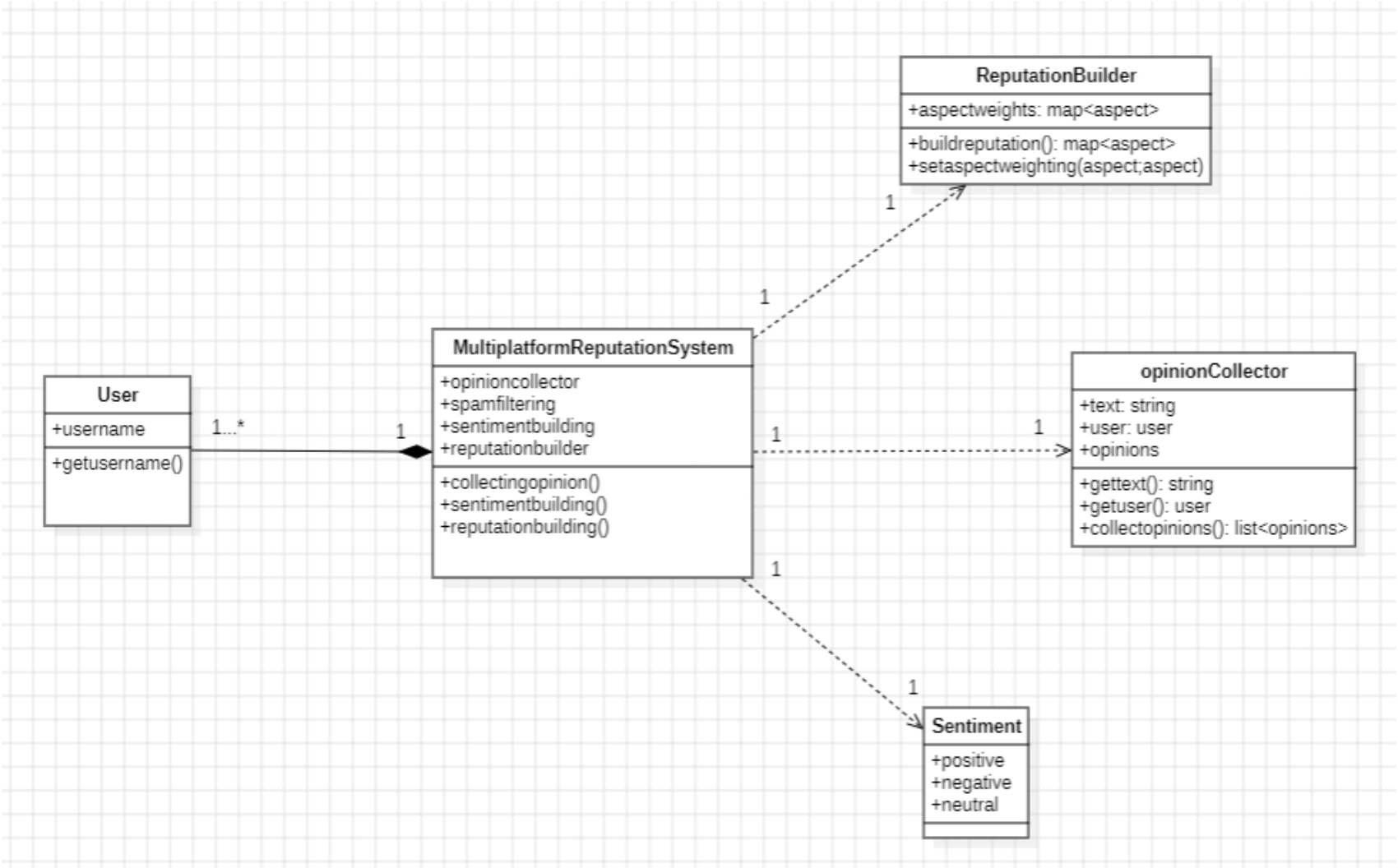
UML

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram

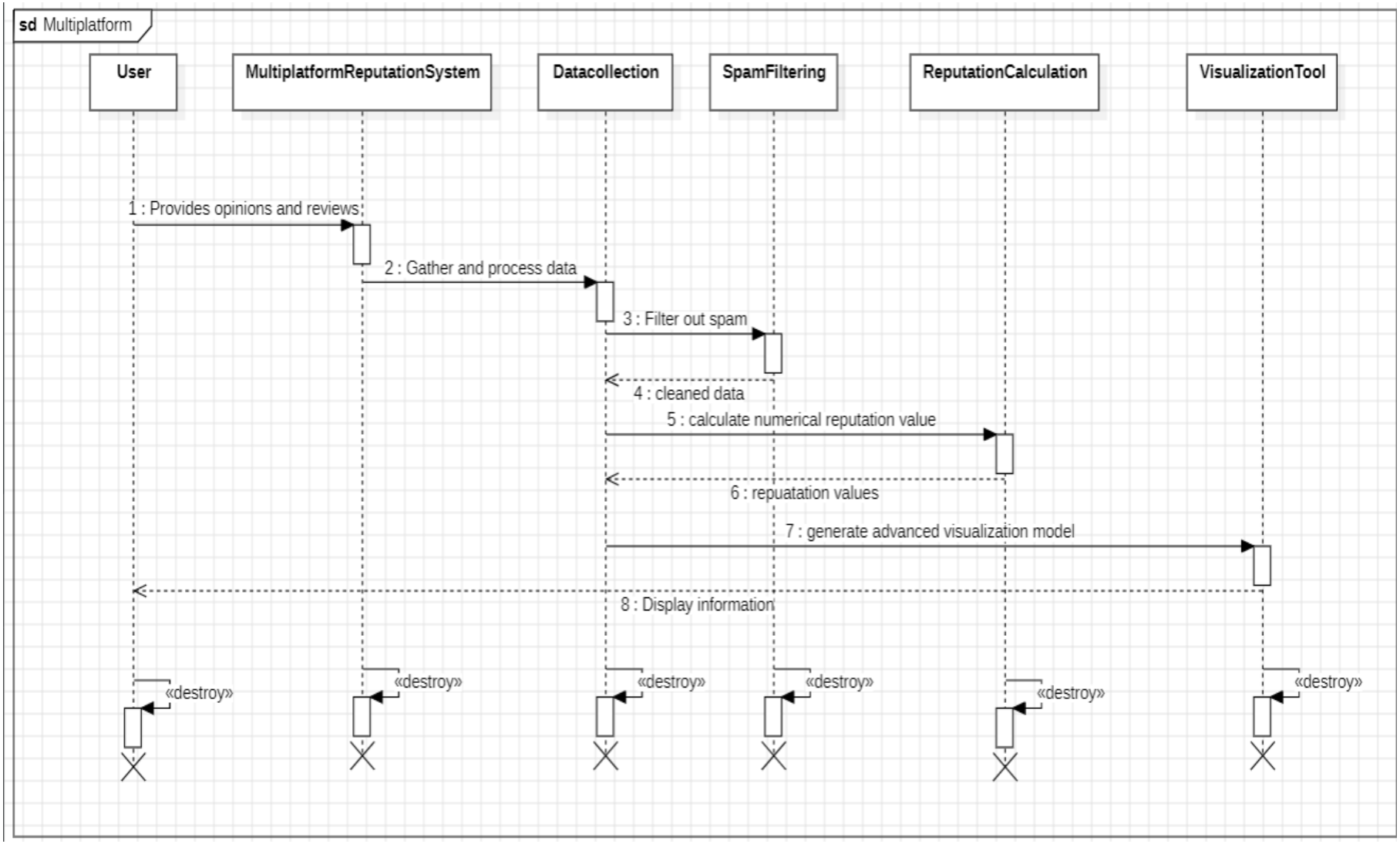
Use Case Diagram



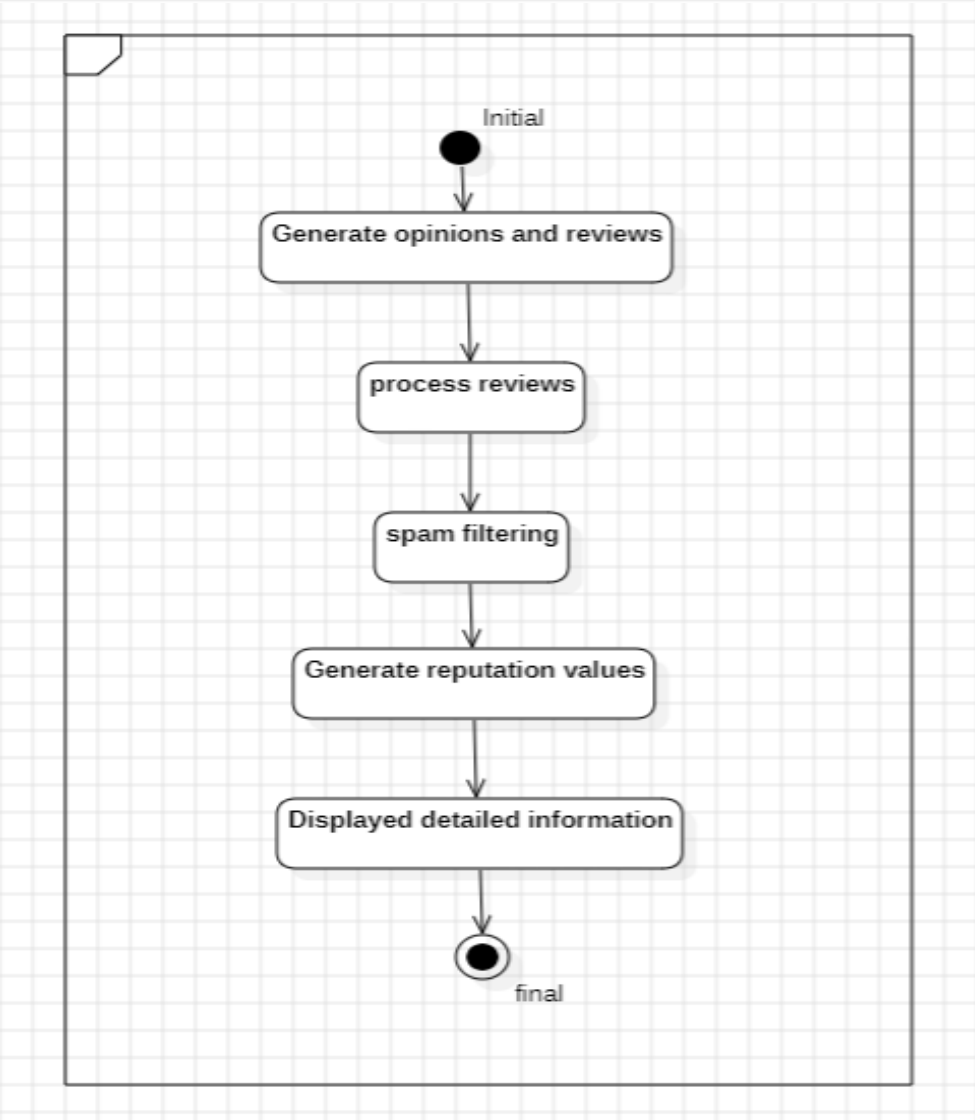
Class Diagram



Sequence Diagram



Activity Diagram



yelp=pd.read_csv('yelp.csv')

Test Cases

Test Case 1:Uploading Dataset

Dataset yelp.csv will be uploaded using yelp=pd.read_csv('yelp.csv')

	business_id	date	review_id	stars	text	type	user_id	cool	useful	funny
0	9yKzy9PApeiPPOUJEtnvkg	2011-01-26	fWKvX83p0-ka4JS3dc6E5A	5	My wife took me here on my birthday for breakf...	review	rLtl8ZkDX5vH5nAx9C3q5Q	2	5	0
1	ZRJwVLyzEJq1VAihDhYiow	2011-07-27	IjZ33sJrzXqU-0X6U8NwyA	5	I have no idea why some people give bad review...	review	0a2KyEL0d3Yb1V6aivbluQ	0	0	0
2	6oRAC4uyJCsJl1X0WZpVSA	2012-06-14	lESLBzqUCLdSzSqm0eCSxQ	4	love the gyro plate. Rice is so good and I als...	review	0hT2KtfLiobPvh6cDC8JQg	0	1	0
3	_1QQZuf4zZOyFCvXc0o6Vg	2010-05-27	G-WvGaISbqqaMHlNnByodA	5	Rosie, Dakota, and I LOVE Chaparral Dog Park!!...	review	uZetl9T0NcROGOyFfughhg	1	2	0
4	6ozycU1RpktNG2-1BroVtw	2012-01-05	1uJFq2r5QfJG_6ExMRCaGw	5	General Manager Scott Petello is a good egg!!!...	review	vYmM4KTsC8ZfQBg-j5MWkw	0	0	0

Remark: Successfully Uploaded

Test Cases

Test Case 2 :Creating a new column text length

Text length will be created using yelp['text length']=yelp['text'].apply(len)

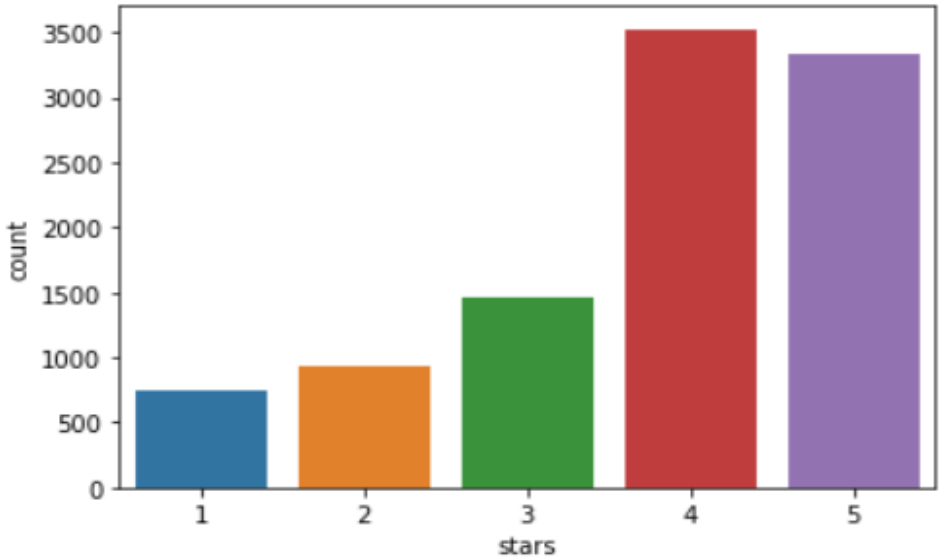
	business_id	date	review_id	stars	text	type	user_id	cool
0	9yKzy9PApeiPPOUJEtnvkg	2011-01-26	fWKvX83p0-ka4JS3dc6E5A	5	My wife took me here on my birthday for breakf...	review	rLtl8ZkDX5vH5nAx9C3q5Q	2
1	ZRJwVLyzEJq1VAihDhYiow	2011-07-27	IjZ33sJrzXqU-0X6U8NwyA	5	I have no idea why some people give bad review...	review	0a2KyEL0d3Yb1V6aivbIuQ	0
2	6oRAC4uyJCsJl1X0WZpVSA	2012-06-14	lESLBzqUCLdSzSqm0eCSxQ	4	love the gyro plate. Rice is so good and I als...	review	0hT2KtfLiobPvh6cDC8JQg	0
3	_1QQZuf4zZOyFCvXc0o6Vg	2010-05-27	G-WvGaISbqqaMHlNnByodA	5	Rosie, Dakota, and I LOVE Chaparral Dog Park!!...	review	uZetl9T0NcROGOyFfughhg	1
4	6ozycU1RpktNG2-1BroVtw	2012-01-05	1uJFq2r5QfJG_6ExMRCaGw	5	General Manager Scott Petello is a good egg!!!...	review	vYmM4KTsC8ZfQBg-j5MWkw	0
...	...	...	...	...	...	...	...	...
9995	VY_tvNUCCXGXQeSvJl757Q	2012-07-28	Ubyfp2RSDYW0g7Mbr8N3iA	3	First visit...Had lunch here today - used my G...	review	_eqQoPtQ3e3UxLE4faT6ow	1
9996	EKzMHI1tip8rC1-ZAy64yg	2012-01-18	2XyIOQKbVFb6uXQdJ0RzlQ	4	Should be called house of deliciousness!\n\nI ...	review	ROru4uk5SaYc3rg8IU7SQw	0

Remark: Successfully created

Test Cases

Test Case 3 :Counting reviews in each rating

```
<matplotlib.axes._subplots.AxesSubplot at 0x1ca5f6cfa48>
```

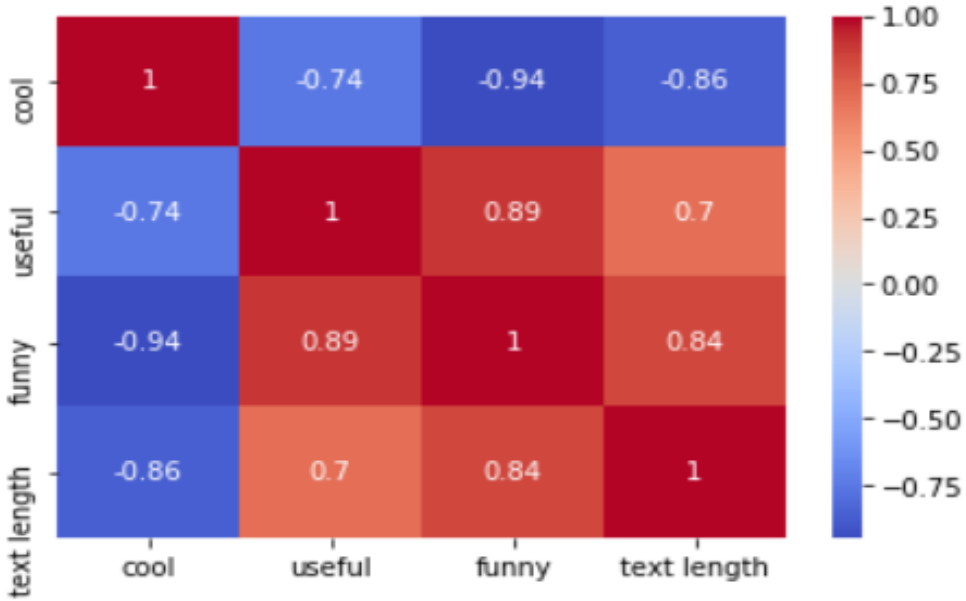


Remark: Successfully Done

Test Cases

Test Case 4 :Checking any correlation between features

```
: <matplotlib.axes._subplots.AxesSubplot at 0x1ca5e30cc88>
```



Remark: Successfully Done

Test Cases

Test Case 5 :Testing with Algorithms

Algorithms are tested by creating a pipeline to make execution simpler.

Remark: Successfully Done

Result

Result-1:when Navie Bayes Algorithm is used

Output:

```
              precision    recall  f1-score   support

           1       0.68      0.91      0.78       170
           5       0.98      0.93      0.96      1056

    accuracy                           0.93      1226
   macro avg       0.83      0.92      0.87      1226
weighted avg       0.94      0.93      0.93      1226

[[155  15]
 [ 73 983]]
```

Result

Result-1:when Navie Bayes Algorithm is used

Output:

Rating	Precision	Recall	F1-Score	Support
1	0.68	0.91	0.78	170
5	0.98	0.93	0.96	1056

Result

Result-2:when Logistic Regression Algorithm is used

Output:

```
              precision    recall  f1-score   support

           1       0.77      0.87      0.82       201
           5       0.97      0.95      0.96      1025

    accuracy                           0.94      1226
   macro avg       0.87      0.91      0.89      1226
weighted avg       0.94      0.94      0.94      1226

[[175  26]
 [ 53 972]]
```

Result

Result-2:when Logistic Regression Algorithm is used

Output:

Rating	Precision	Recall	F1-Score	Support
1	0.77	0.87	0.82	201
5	0.97	0.95	0.96	1025

Result

Result-2:when SVM Regression Algorithm is used

Output:

```
Classification Report:
              precision    recall  f1-score   support

           1       0.36      0.98      0.53        84
           5       1.00      0.87      0.93      1142

    accuracy                           0.88      1226
   macro avg       0.68      0.92      0.73      1226
weighted avg       0.95      0.88      0.90      1226


Confusion Matrix:
[[ 82   2]
 [146 996]]
```

Result

Result-2:when SVM Regression Algorithm is used

Output:

Rating	Precision	Recall	F1-Score	Support
1	0.36	0.98	0.53	84
5	1.00	0.87	0.93	1142

Result

Result-2:when K-Nearest Neighbor Algorithm is used

Output:

```
              precision    recall  f1-score   support

           1       0.09      0.72      0.16        29
           5       0.99      0.83      0.90      1197

    accuracy                           0.82      1226
   macro avg       0.54      0.78      0.53      1226
weighted avg       0.97      0.82      0.88      1226

[[ 21   8]
 [207 990]]
```

Result

Result-2:when K-Nearest Neighbor Algorithm is used

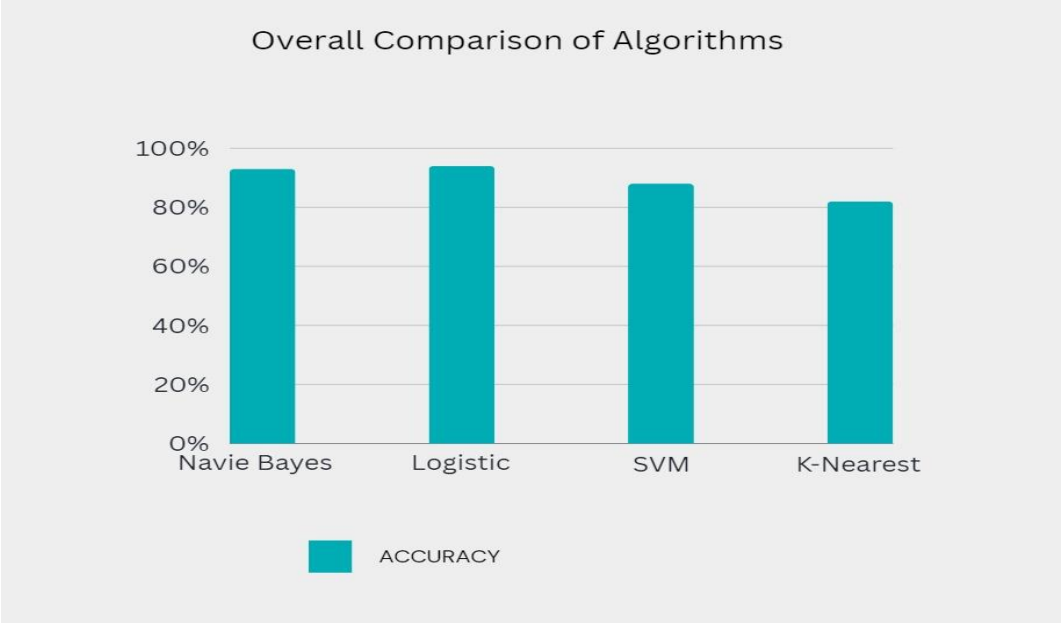
Output:

Rating	Precision	Recall	F1-Score	Support
1	0.09	0.72	0.16	29
5	0.99	0.83	0.90	1197

Result

Comparative Analysis of Algorithm:

- All the models gave accuracy with Naive Bayes at 93% , Logistic Regression at 94% , SVM Algorithm at 88% and K-nearest Algorithm at 82%
- Though Logistic Regression has overall better precision.



Result

Comparative Analysis of Algorithm:

Algorithm	Accuracy
Navie Bayes Algorithm	0.93
Logistic Algorithm	0.94
SVM Algorithm	0.88
K-Nearest Neighbor Algorithm	0.82

- the accuracy values for the various algorithms. From the above table we can depict that logistic Algorithm gives the highest accuracy values so we can use this algorithm to predict the sentiment of the model.
- Finally, By using Logistic Algorithm we can predict the sentiment of the model by using confusion matrix that ,the model has positive reputation .

FUTURE SCOPE

- Looking ahead, the future scope of our innovative reputation system entails continual refinement and expansion. We aim to enhance its capabilities by incorporating advanced machine learning techniques for more nuanced sentiment analysis.
- Additionally, the system will evolve to generate concise textual summaries of entity attributes, providing users with even more valuable insights. Recognizing the global nature of online content, we plan to adapt the system for seamless operation in multilingual environments.
- Continuous research and development will ensure that our reputation system remains at the forefront, catering to the evolving needs of diverse users and staying abreast of emerging technologies in the field of sentiment analysis and opinion evaluation.

Conclusion

We introduced a novel reputation system for assessing items and their aspects from online opinions. Our system stands out by being Multiplatform compatible, gathering and standardizing opinions from diverse sources. It employs advanced spam filtering, utilizes a state-of-the-art aspect-based sentiment-analysis model. The resulting reputation values are presented through a comprehensive visualization. In user and expert evaluations, our system outperformed four state-of-the-art systems. Future plans include expanding the system to generate textual summaries of entity attributes and adapting it for use in multilingual content.

THANK YOU